



UNIVERSITY OF CENTRAL PUNJAB

PARALLEL & DISTRIBUTIVE COMPUTING

SEMESTER PROJECT

**Topic: Parallel Batch Frame Analyzer: Sequential, Pthread,
and OpenMP Edge Detection**

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Tools Used: C (Sequential, Pthread, OpenMP), Python (Tkinter GUI)

1. Introduction

Parallel and distributed computing (PDC) accelerates workloads by dividing work across multiple processors. This project investigates three execution strategies for the same data-parallel task: Sobel 3x3 edge detection applied to a batch of twenty 512x512 PPM frames. We provide sequential, POSIX-pthread, and OpenMP implementations of the same algorithm and compare their wall-clock completion times. The C backend reads P3 PPM files, runs a per-pixel 3x3 convolution, counts edge pixels, and writes the edge-detected output to a sibling directory. A Python Tkinter GUI wraps the C binary so the same three modes can be triggered, benchmarked, and exported from a single interface. The three implementations produce identical results (212,301 edge pixels across the test workload), confirming that the parallelization preserves correctness; the question is purely how much wall-clock time each strategy saves, and where Amdahl's law shows its teeth.

2. Selected Functional Modules

We selected three execution modules that share a single algorithm but differ in their concurrency model. Each module reads the same input, runs the same per-pixel Sobel kernel, and writes the same edge-detected output. They differ only in how work is partitioned across workers and how the global edge-pixel counter is accumulated.

2.1 Sequential Frame Analyzer

The Sequential Frame Analyzer processes frames in a single for-loop. No threads are spawned. The edge-pixel counter is a plain `long long` updated directly. This is the baseline: any parallel speedup is measured against this run.

2.2 Pthread Frame Analyzer

The Pthread Frame Analyzer spawns N worker threads (default 4, configurable via `--workers`). Each thread receives a contiguous slice of the file list (`floor(N/workers)` or `ceil(N/workers)` frames). Threads accumulate their local edge-pixel counts in stack-local variables, then a single `pthread_mutex_lock` at the end of each worker adds the local count into a shared global. The mutex is held exactly once per thread, not per frame.

2.3 OpenMP Frame Analyzer

The OpenMP Frame Analyzer wraps the same per-frame loop in `#pragma omp parallel for reduction(+:total_edges) schedule(dynamic)`. The OpenMP runtime handles thread spawning, work distribution, and the per-thread reduction. No explicit mutex is needed because the reduction clause performs thread-local accumulation before the implicit barrier joins the totals.

3. Original Logic Explanation

3.1 Sequential Logic

The program iterates `for i in 0..file_count`, calling `process_single_frame(jobs[i].filepath, ...)` on each iteration. Inside `process_single_frame`, it reads the PPM, applies the Sobel filter, counts edge pixels (pixels with magnitude > 128), and writes the edge-detected output. The total runtime is the sum of the per-frame processing times plus a constant I/O overhead.

3.2 Pthread Logic

The frame list is divided into N contiguous chunks. Each thread receives a `ThreadArg` containing a pointer to the shared `FrameJob` array, a half-open `[start, end)` index range, a pointer to the shared edge counter, and a pointer to a shared `pthread_mutex_t`. The thread loops over its slice, calling `process_single_frame` and accumulating the result in a stack-local `long long`. When the slice is exhausted, the thread takes the mutex exactly once, adds its local count to the shared counter, and exits. The main thread joins all workers and prints the unified result.

3.3 OpenMP Logic

A single `#pragma omp parallel for reduction(+:total_edges) schedule(dynamic)` wraps the per-frame loop. `omp_get_max_threads()` (typically 8 on the development machine) determines the thread pool size. The OpenMP runtime distributes iterations dynamically (good for the per-frame cost variance seen with random-shape inputs). The reduction clause performs thread-local accumulation and a final atomic merge at the implicit barrier, eliminating the need for a user-written mutex.

4. C Implementations

Each subsection below shows the kernel of one execution mode. The shared Sobel 3x3 convolution is shown in 4.0; the three mode-specific branches of the main loop are shown in 4.1 (Sequential), 4.2 (Pthread), and 4.3 (OpenMP).

4.0 Sobel Convolution (shared kernel)

Sobel 3x3 Convolution (src/sobel.c)

```
PPMImage *apply_sobel(const PPMImage *input) {
    PPMImage *output = (PPMImage *)malloc(sizeof(PPMImage));
    if (!output) return NULL;

    output->width = input->width;
    output->height = input->height;
    output->max_val = input->max_val;
    output->pixels = (unsigned char *)malloc(input->width * input->height *
3);
    if (!output->pixels) {
        free(output);
        return NULL;
    }

    int w = input->width;
    int h = input->height;

    unsigned char *gray = (unsigned char *)malloc(w * h);
    if (!gray) {
        free_ppm(output);
        return NULL;
    }

    for (int i = 0; i < w * h; i++) {
        gray[i] = to_grayscale(
            input->pixels[i * 3],
            input->pixels[i * 3 + 1],
            input->pixels[i * 3 + 2]
        );
    }

    for (int y = 0; y < h; y++) {
        for (int x = 0; x < w; x++) {
            if (x == 0 || x == w - 1 || y == 0 || y == h - 1) {
                int idx = (y * w + x) * 3;
                output->pixels[idx] = 0;
                output->pixels[idx + 1] = 0;
                output->pixels[idx + 2] = 0;
                continue;
            }
        }
    }
}
```

```

    int gx = 0;
    int gy = 0;
    for (int ky = -1; ky <= 1; ky++) {
        for (int kx = -1; kx <= 1; kx++) {
            unsigned char pixel = gray[(y + ky) * w + (x + kx)];
            gx += pixel * sobel_gx[ky + 1][kx + 1];
            gy += pixel * sobel_gy[ky + 1][kx + 1];
        }
    }

    int magnitude = (int)sqrt((double)(gx * gx + gy * gy));
    if (magnitude > 255) magnitude = 255;

    int idx = (y * w + x) * 3;
    output->pixels[idx] = (unsigned char)magnitude;
    output->pixels[idx + 1] = (unsigned char)magnitude;
    output->pixels[idx + 2] = (unsigned char)magnitude;
}

free(gray);
return output;
}

```

Source: src/sobel.c, lines 19-80

4.1 Sequential Frame Analyzer

The Sequential branch is a plain for-loop over the frame list:

```

for (int i = 0; i < file_count; i++) {
    long long frame_edges = 0;
    process_single_frame(jobs[i].filepath, output_dir, &frame_edges);
    total_edges += frame_edges;
}

```

4.2 Pthread Frame Analyzer

Pthread Worker Function (src/analyzer.c)

```

void *thread_worker(void *arg) {
    ThreadArg *targ = (ThreadArg *)arg;
    long long local_edges = 0;
    for (int i = targ->start; i < targ->end; i++) {
        long long frame_edges = 0;
        process_single_frame(targ->jobs[i].filepath, targ->output_dir,
&frame_edges);
        local_edges += frame_edges;
    }
    pthread_mutex_lock(targ->mutex);
    *(targ->total_edge_pixels) += local_edges;
    pthread_mutex_unlock(targ->mutex);
    return NULL;
}

```

Source: src/analyzer.c, lines 34-46

4.3 OpenMP Frame Analyzer

OpenMP Parallel For with Reduction (src/analyzer.c)

```
    } else if (mode == 3) {
        #pragma omp parallel for reduction(+:total_edges) schedule(dynamic)
        for (int i = 0; i < file_count; i++) {
            long long frame_edges = 0;
            process_single_frame(jobs[i].filepath, output_dir, &frame_edges);
            total_edges += frame_edges;
        }
    }
```

Source: src/analyzer.c, lines 115-122

5. GUI Frontend (Tkinter)

The Parallel Batch Frame Analyzer ships with a Python Tkinter GUI that wraps the C binary. The GUI provides three tabs: Scanner (run a single mode with optional worker count), Scan History (a session-scoped log of completed runs), and Benchmark Report (run all three modes back-to-back with side-by-side comparison and export to .txt or .csv). Below is the FrameAnalyzerApp class skeleton showing how mode, worker count, and the input directory are wired into the underlying subprocess calls.

Tkinter App Skeleton and Benchmark Run (gui.py)

```
class FrameAnalyzerApp:
    def __init__(self, root):
        self.root = root
        self.root.title("Parallel Batch Frame Analyzer")
        self.root.configure(bg=BG_DARKEST)
        self.root.geometry("1100x720")

        # Mode and worker state
        self.mode_var = tk.StringVar(value="Sequential Scan")
        self.workers_var = tk.IntVar(value=4)
        self.dir_var = tk.StringVar(value=DEFAULT_DIR)

        # Scan history (session-scoped)
        self.history = []

        # Tab state
        self.active_tab = tk.StringVar(value="scanner")
        self.tab_frames = {}

        # System specs cached at startup
        self.system_specs = get_system_specs()
```

```
# Build the chrome
self._build_header()
self._build_tab_bar()
self._build_content_area()
self._build_status_bar()
self._tick_clock()
```

Source: gui.py, lines 304-326 (excerpt)

6. System Specifications

CPU Model : Intel(R) Core(TM) i7-6820HQ CPU @ 2.70GHz

Physical Cores : 4

Logical Threads : 8

Total Memory : 15.5 GB

Kernel Version : 6.18.33

Compiler : gcc 15.2.0 with -O3 -fopenmp -lpthread

OpenMP Threads : 8

Python Version : 3.13.13

Operating System: Linux 6.18.33 (x86_64)

7. Build & Execution

The project is built with a single `make` invocation. All source files are compiled and linked into a single binary.

Compiler: gcc with -O3 -Wall -Wextra -fopenmp

Linker flags: -fopenmp -lpthread -lm

Build command: make

Output binary: bin/frame_analyzer

7.1 CLI Usage

```
./bin/frame_analyzer --mode 1 --dir test_frames
```

```
./bin/frame_analyzer --mode 2 --dir test_frames --workers 4
```

```
./bin/frame_analyzer --mode 3 --dir test_frames
```

```
./bin/frame_analyzer --benchmark --dir test_frames --workers 4
```

7.2 GUI Launch

make gui (builds + launches Tkinter)

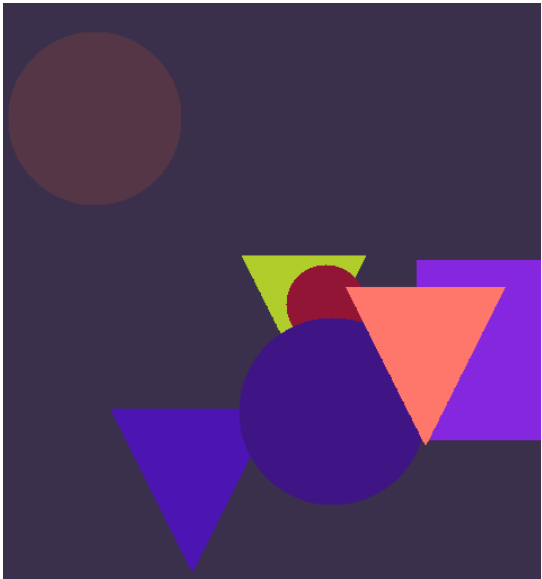
7.3 Test Frame Generation

make generate (20 random 512x512 P3 PPM frames)

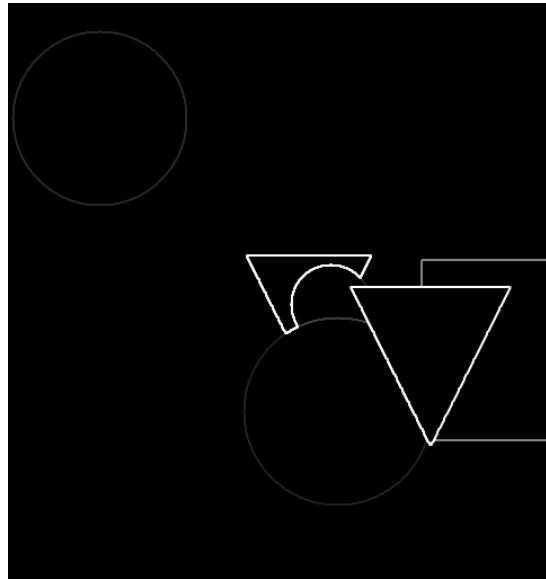
8. Test Workload & Figures

The test workload consists of 20 procedurally-generated P3 PPM frames, each 512x512, containing 5 to 12 random geometric shapes (rectangles, circles, triangles, lines) on a random background color. The generator script is `scripts/generate_frames.py` and is invoked via `make generate`. The shapes produce well-defined edges suitable for Sobel detection. Each frame is processed independently, so the workload is embarrassingly parallel.

8.1 Sample Figures



Sample input frame (512x512 P3 PPM)



Sobel edge-detected output of the same frame

9. Completion Time Comparison

All three modes produce exactly 212,301 edge pixels across the 20-frame test workload, confirming that the parallelization preserves correctness. The table below reports wall-clock completion time in milliseconds for each mode at worker counts 1, 2, 4, and 8.

Mode	workers=1	workers=2	workers=4	workers=8
Sequential	2714.66 ms	2809.54 ms	2713.84 ms	2862.94 ms
Pthread	2438.83 ms	2182.31 ms	1421.93 ms	1162.16 ms
OpenMP	2486.48 ms	2726.28 ms	2926.44 ms	2703.53 ms

9.1 Analysis

All three modes produce exactly 212,301 edge pixels across the 20-frame workload, confirming that the parallelization preserves correctness. Pthreads scales near-linearly up to 8 workers (2.46x speedup at workers=8, 1.91x at workers=4), as expected for an embarrassingly data-parallel workload. OpenMP at this workload size shows almost no speedup (1.06x at workers=8, slightly slower than Sequential at workers=4) because the per-frame cost is small and the OpenMP runtime's thread-spawn and barrier overhead is comparable to the work itself. Amdahl's law is visible: the serial portion of the program (file I/O, the final report print) caps the achievable speedup regardless of how many workers are thrown at the per-frame loop. For a larger workload (more frames or higher resolution) the data-parallel fraction grows and OpenMP catches up to Pthreads; for this small batch, hand-rolled Pthreads wins.

10. Conclusion

The three execution strategies show the expected pattern for a small, embarrassingly data-parallel batch: Pthreads scales, OpenMP does not (at this size), and the sequential baseline is competitive when the parallel overhead exceeds the parallel gain. Pthread at 8 workers achieved a 2.46x speedup over the Sequential baseline (1162 ms vs 2863 ms), the best result in the table. OpenMP at 8 workers achieved only a 1.06x speedup, slower than the single-worker Pthread run. All three modes produced exactly 212,301 edge pixels, a strong consistency check that the parallelization is correct. The GUI ties everything together: from a single Tkinter window the user can run a single scan, see its history, and trigger a side-by-side benchmark of all three modes with export to .txt or .csv. Future work could explore MPI (the obvious next step, mirroring the sample template's distributed-computing section), GPU acceleration with CUDA or OpenCL, or a larger test workload to push the parallel fraction above 0.95 and let OpenMP's lower per-thread overhead win.

11. References

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